

Example Analysis of Oak Woodlands

Example of an Ecological Data Set

Oak Woodlands in the Willamette Valley, Oregon, USA

In 1961 and 1962 John F. Thilenius sampled vascular plants in oak forests in the Willamette Valley for his Ph.D. at Oregon State University (Thilenius 1963, 1968). The data came from a fairly narrow range of habitats – all of the stands were closed forests dominated by *Quercus garryana*. This resulted in a data set with fairly low beta diversity. The environmental differences among the sites are rather modest. Much of the variation in species composition presumably is derived from the particular histories of each stand, such as episodes of grazing, logging, and fire. Of course we have limited information on those histories, so you will see that much of the variation in the plant communities is not readily explained by the measured environmental and historical variables. Nevertheless a definite environmental gradient emerges from the analysis.

The abstract from Thilenius (1968) is reproduced below:

"*Quercus garryana* forests, prominent at low elevations throughout the Willamette Valley, Oregon, have developed from oak savanna subsequent to settlement of the valley in the mid-nineteenth century. Interruption of the ground fires that were common in the pre-settlement environment probably caused the change. The understory of the oak forest is dominated by shrubs, and well-defined strata are present. Four plant communities occur: (1) *Quercus garryana*/*Corylus cornuta* var. *californica*/*Polystichum munitum* (most mesic); (2) *Quercus garryana*/*Prunus avium*/*Symphoricarpos albus*; (3) *Quercus garryana*/*Amelanchier alnifolia*; (4) *Quercus garryana*/*Rhus diversiloba* (most xeric). All are in seral condition because of their relatively recent development and because they have been disturbed throughout their existence by man's activities. The soils supporting the oak forest are generally deep and well drained and have developed profiles with illuvial horizons and acidic reaction. They are derived from sedimentary and basic igneous rocks and old valley-filling alluvium. Seven established soil series are present: Steiwer, Carlton, Peavine, Nekia, Dixonville, Olympic, and Amity. The Steiwer series and its catenary associate, Carlton, are the most common soils."

Thilenius' goals were to describe "the floristic composition, stand structure, physical environment, and successional status of plant communities where *Quercus garryana* is the major component of the overstory." Although quantitative data were carefully recorded, Thilenius had few possibilities for multivariate analysis. His primary analyses were first arranging his data "according to similarities in species composition, importance ranks, and environmental attributes." He then tabulated averages for species and environmental variables within the four groups.

Here is an interesting challenge for modern community analysts: what can you add to his account (Thilenius 1968) based on a more sophisticated quantitative analysis of the data? I mentioned above that a single strong environmental gradient emerges from the analysis, but this is only hinted in Thilenius' abstract. What is that gradient?

After a listing of the files and variables contained in the files, three example procedures are given. The first demonstrates modification of the raw data into a form suitable for analysis. The second is an ordination with nonmetric multidimensional scaling. The third compares groups of sample units, as defined by landform.

Files Provided

OakWoods.doc – Microsoft Word file containing this document.

OakWood1.mjm – Main matrix containing species abundances in a matrix of 47 stands x 103 species. Abundances were derived from basal areas for trees and canopy cover for other species. Some species are listed more than once because abundance was evaluated separately for different height classes, as indicated by the suffixes: t = tree, s = shrub. Species with fewer than three occurrences were deleted, then the matrix was relativized by species maximum (i.e. each element in the matrix is expressed as a proportion of the maximum value in each column). Relativization by columns is necessary for some analyses because different columns were measured in different units. See OakRaw.mjm for the raw data.

OakWood2.mjm – Second matrix of 47 stands x 27 attributes. The attributes are described in detail below. They include environmental variables, indicators of stand history. I have also added some community summary variables, including species richness, groups derived from cluster analysis, and community types as originally designated by Thilenius.

OakRaw.mjm – The raw data matrix containing 47 stands x 189 species, before any modifications. The values are basal areas (ft^2/acre) for trees and percentage cover for lower strata, based on $60, 0.2 \text{ m}^2$ quadrats/stand. "Trace" was converted to 0.5%. A check on the field data sheet was converted to 0.2%. Be careful! Any use of these raw data must recognize that the columns representing the tree stratum differ in units from the lower strata; hence, use of a relativized matrix in OakWood1.mjm.

More on the methods from Thilenius (1968):

"Investigations were confined to closed-canopy stands 4 ha or more in area where *Quercus garryana* was the major component of the overstory. Basal area, frequency, and density of overstory trees were determined on twenty 0.004-ha circular plots spaced at 9-m intervals in four rows parallel to the slope contour. Density was recorded in four classes: saplings (< 10 cm dbh); poles (11-40 cm dbh); mature (41-100 cm dbh) and relict (> 100 cm dbh). The maximum height of trees on each plot was measured with an optical rangefinder."

"Frequency and percentage crown coverage of shrub and herbaceous species were recorded on sixty 0.2 m^2 quadrats spaced at 3-m intervals in four rows coincident with the rows of 0.004-ha plots. Very low crown coverage was recorded as trace and arbitrarily assigned a value of 0.5% for calculation purposes. Above trace, the intervals were 1% and 5%. Coverage greater than 5% was estimated to the nearest 10%."

Coding for Variables in the Second Matrix

Topographic and geographic variables

Elev.m = elevation above sea level in meters.

LatAppx = approximate latitude, decimal degrees, based on automated conversion of Township/Range/Section, using the program TRS2LL.exe.

LongAppx = approximate longitude, decimal degrees, based on automated conversion of Township/Range/Section, using the program TRS2LL.exe.

SlopeDeg = slope in degrees (originally recorded in percentages)

AspClass = aspect class, 1=SW, 2=S or W, 3=SE or NW, 4=N or E, 5=NE.

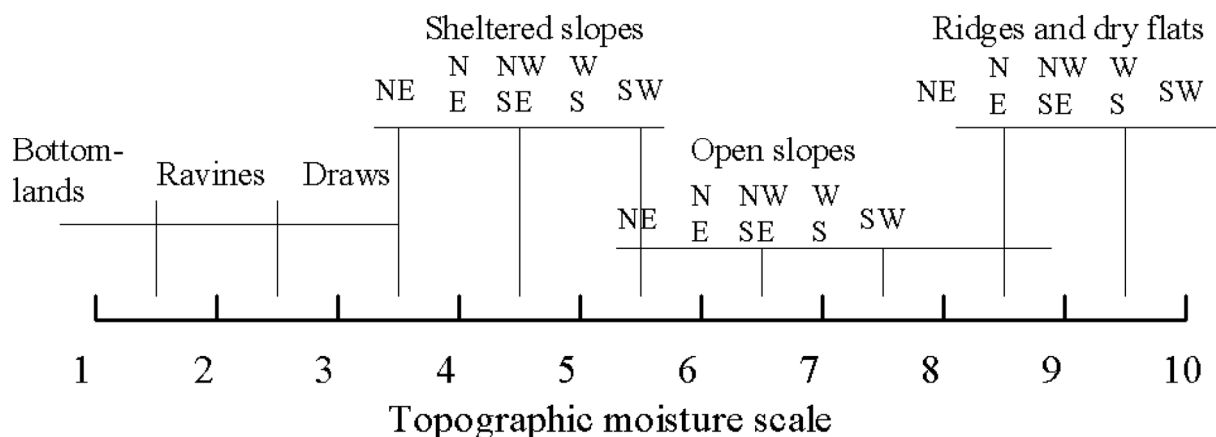
AspDeg = aspect in degrees E of N

PDIR = Potential annual direct incident radiation, $\text{MJ}/\text{cm}^2/\text{yr}$, calculated according to McCune and Keon (2002) Eq. 3.

HeatLoad = Heat load index, calculated according to McCune and Keon (2002).

Landform: 1=valley bottom, 2=draw or slope of draw, 3=slope, 4=ridge

TopoClas = Topographic position class: adapted from scales used by Whittaker & Kessell (Kessell 1979):



Soil variables

Drainage: 1=poor, 2=moderate, 3=good, 4=well.

Soil series: 1=Steiwer, 2=Peavine, 3=Dixonville, 4=Nekia, 5=Carlton, 6=Olympia, 7=Amity

SoilGrp: 1=sedimentary, 2=basic igneous, 3=alluvial

A-horiz = thickness of A horizon, cm

B1-horiz = thickness of B1 horizon, cm

B2-horiz = thickness of B2 horizon, cm

B3-horiz = thickness of B3 horizon, cm (if profile truncated, e.g. "44+ inches", add 20 inches)

B-horiz = sum of B1+B2+B3, cm

Indicators of stand history

GrazCurr = signs of current grazing recorded on field data sheet (0=no, 1=yes)

GrazPast = signs of past grazing recorded on field data sheet (0=no, 1=yes, must be 1 if GrazCurr=1)

NotLogged = NPL recorded under "Influences" on data sheet. I guessed this means "no past logging", i.e. no signs of past logging.

Que>60cm = number of *Quercus garryana* recorded in the 60 cm (24 inch) size class and larger. (No stands had large *Pseudotsuga*; one stand (Stand05) had a large *Acer macrophyllum* and one stand (Stand07) had two large *Arbutus menziesii*)

LogQ>60 = log of (x+1) where x is the number of *Quercus garryana* recorded in the 60 cm (24 inch) size class and larger (i.e. x = "LogQ>60").

TreeHm = maximum height of *Quercus garryana* in meters.

Community summary variables derived from the species matrix

SppRich = species richness, calculated from OakRaw.mjm, counting each species x layer combination as a separate species.

ThlType = vegetation types from Thilenius (1968)

1 = *Quercus/Corylus/Polystichum*

2 = *Quercus/Prunus/Symphoricarpos*

3 = *Quercus/Amelanchier/Symphoricarpos*

4 = *Quercus/Rhus*

FxB-.25 = community types defined at the 4-group level from hierarchical cluster analysis, Flexible beta method, Serensen distance, beta= -.025.

List of Species Codes

Note: because woody species may occur in more than one stratum, a suffix (-s, -t) is used to indicate a given species in the shrub or tree stratum.

Abgr#s Abies grandis SHRUB
 Abgr-t Abies grandis
 Acar Actaea arguta
 Acgld Acer glabrum var. douglasii
 Acma#s Acer macrophyllum shrub
 Acma-t Acer macrophyllum
 Acmi Achillea millefolium
 Adbi Adenocaulon bicolor
 Agha Agrostis hallii
 Agre Agropyron repens
 AGRO Agrostis sp?
 Agse Agrostis semiverticillata (subsecundum)
 Agte Agrostis tenuis
 Aica Aira caryophylla
 ALL Allium sp.
 Alpr Alopecurus pratensis
 Amal#s Amelanchier alnifolia shrub
 Amal-t Amelanchier alnifolia
 Apan Apocynum androsaemifolium
 Aqfo Aquilegia formosa
 Arel Arrhenatherum elatius
 Arme#s Arbutus menziesii SHRUB
 Arme-t Arbutus menziesii
 Avfa Avena fatua
 Beaq Berberis aquifolium
 Brpu Brodiaea pulchella
 Brco Bromus commutatus
 Brla Bromus laevipes
 Brri Bromus rigidus
 Brse Bromus secalinus
 Brst Bromus sterilis
 Brvu Bromus vulgaris
 Caqu Camassia quamash
 CAR Carex sp.
 Cato Calochortus tolmiei
 Cear Cerastium arenses
 Ceum Centaurium umbellatum
 Ceve Ceanothus velutinus
 Cipa Circaea pacifica
 Civu Cirsium vulgare
 Coco#s Corylus cornuta shrub
 Coco-t Corylus cornuta
 Cogr Collomia grandiflora
 Conu#s Cornus nuttallii SHRUB
 Conu-t Cornus nuttallii
 CORY Corylus sp.
 Cost Corallorhiza striata
 Crca Crepis capillaris
 Crdo#t Crataegus douglasii
 Crdo#s Crataegus douglasii
 Crox Crataegus oxyacantha
 Cyec Cynosurus echinatus
 Cyfo Cystopteris fragilis
 Cygr Cynoglossum grande
 Daca Danthonia californica
 Dacar Daucus carota
 Dagl Dactylus glomerata
 Deel Deschampsia elongata
 Diar Dianthus armeria
 Doel Downingia elegans
 Drar Dryopteris arguta
 Drar Dryopteris arguta
 Elgl Elymus glaucus
 Eria Eriophyllum lanatum
 Erog Erythronium oregonum
 Eucr Euphorbia crenulata
 Feca Festuca californica
 Fede Festuca dertonenses
 Feel Festuca elatior var. arendmaceae
 Feme Festuca megalura
 Feoc Festuca occidentalis
 Feru Festuca rubra
 Frbr Fragaria bracteata (vesca)
 Frcu Fragaria cuneifolia
 Frla#s Fraxinus latifolia shrub
 Frla-t Fraxinus latifolia
 Frvi Fragaria virginiana
 GAL Galium sp.
 Gema Geum macrophyllum
 Geog Geranium oreganum (incisum)
 Gepu Geranium pusillum
 Haob Habenaria orbiculata
 Haun Habenaria unalacensis
 Hehe Hedera helix
 Hemi Heuchera micrantha
 Hodi Holodiscus discolor
 Hola Holcus lanatus
 Hyoc Hydrophyllum occidentale
 Hype Hypericum perforatum
 Hyra Hypochaeris radicata
 Irte Iris tenax
 JUNC Juncus sp.
 Kocr Koeleria cristata
 Laco Lapsana communis
 Lapo Lathyrus polyphyllus
 Lasa Lathyrus sativus (Pisum sativum)
 Liap Ligusticum apiifolium
 Libu Lithophragma bulbifera
 Lico Lillium columbianum
 Lide-t Libocedrus decurrens
 Lide#s Libocedrus decurrens

LILI Liliun sp.
 Loci Lonicera ciliosa
 Lope Lolium perenne
 LOT Lotus sp.
 Lotr Lomatium triternatum
 Lumu Luzula multiflora
 Maex Madia exigua
 MAL Malvaceae sp.
 Maor Mazah oregonus
 Mebu Melica bulbosa
 Mila Microseris laciniata
 Mope Montia perfoliata
 Mosi Montia sibirica
 Nepa Nemophylla parviflora
 ONGR Onagraceae sp.
 Osce#t Osmaronia cerasiformis tree
 Osce-s Osmaronia cerasiformis
 Osch Osmorhiza chilensis
 Osnu Osmorhiza nuda (chilensis)
 Phca Physocarpus capitatus
 Phle Philadelphus lewisii
 Phpr Phleum pratense
 Phvi Phoradendron villosum
 Pipo#s Pinus ponderosa
 Pipo Pinus ponderosa
 Plla Plantago lanceolata
 POCO Poa compressa
 Pogl Potentilla glandulosa
 Pogr Potentilla gracilis
 Pogr Potentilla gracilis
 Pomu Polystichum munitum
 Popr Poa pratensis
 Povu Polypodium vulgare
 Prav#s Prunus avium shrub
 Prav-t Prunus avium
 Prde-t Prunus virginiana var. demissa
 Prde#s Prunus virginiana var. demissa shrub
 Prvu Prunella vulgaris
 Psme#s Pseudotsuga menziesii shrub
 Psme-t Pseudotsuga menziesii
 Ptan Pterospora andromedea
 Ptaq Pteridium aquilinum var. lanuginosum
 Pyco#s Pyrus communis shrub
 Pyco Pyrus communis
 Pyfu#s Pyrus fusca SHRUB
 Pyfu-t Pyrus fusca
 Quqa-s Quercus garryana shrub
 Quqa Quercus garryana
 Raoc Ranunculus occidentalis
 Rhdi Rhus diversiloba
 Rhpu#s Rhamnus purshiana shrub
 Rhpu Rhamnus purshiana
 Risa Ribes sanguinifolius
 Rodu Rosa???
 Roeg Rosa eglanteria
 Rogy Rosa gymnocarpa
 Ronu Rosa nutkana
 Ropi Rosa pisocarpa
 Ropi Rosa pisocarpa
 Ruac Rumex acetosella
 Rula Rubus laciniatus
 Rule Rubus leucodermis
 Rupa Rubus parvifloris
 Rupr Rubus procerus
 Ruur Rubus ursinus
 S#2 Carex sp2.
 S#1 Carex spl.
 Sacr Sanicula crassicaulis
 Sado Satureja douglasii
 Sagr Sanicula graveolens
 Seja Senecio jacobaea
 Siho Silene hookeri
 Smra Smilacina racemosa
 Smse Smilacina sessilifolia
 Syal Symphoricarpos albus
 Taas Taeniatherum asperum
 Taof Taraxacum officinale
 Tegr Tellima grandiflora
 Thoc Thalictrum occidentale
 Toar Torilis arvensis
 Trca Trisetum canescens
 TRIF Trifolium sp
 Trla Trientalis latifolia
 Trov Trillium ovatum
 Trpr Trifolium procumbens
 V#1 Vicia sp.
 Valo Valerianella locusta
 Viam Vicia americana
 Viel Viburnum ellipticum
 Vinu Viola nuttallii
 VIOL Viola sp
 Zice Zygadenus venosus

EXAMPLE ANALYSES

Derivation of an adjusted data matrix (OakWood1.mjm) from the raw data matrix (OakRaw.mjm).

(Note: for more on the rationale behind these steps, see [McCune & Grace \(2002\)](#), "Analysis of Ecological Communities.")

1. Open the file OakRaw.mjm as the main matrix (*File | Open | Main matrix*).
2. Delete species with fewer than three occurrences (*Modify Data | Delete Columns | Fewer than N Non-zero Values | select N=3* (The rationale for this is explained by [McCune & Grace 2002](#), pp. 75-76)
3. Click **OK** in answer to *Do you wish to use Temp.mjm as the new Main Matrix?*
4. Note that the result file now shows a list of the 86 columns (species) that were deleted.
5. *Modify Data | Relativizations | Main matrix | Relativization by maximum* | select *Columns: Species* | **OK**. (It is essential to relativize by columns (species) in this case, because some species have abundances as basal areas and some as percent cover; see p. 73 in McCune & Grace 2002).
6. You will be asked, *Current temporary RESULT.TXT file will be lost. Save file now?* Click on *Discard* (you do not normally need to save this, as it just has the list of the species that were deleted.)
7. Click **OK** in answer to *Do you wish to use Temp.mjm as the new Main Matrix?*
8. You will be asked, *Current temporary work file WORK.MJM will be lost. Save file now?* Click on *Discard* – no need to save this file. It contains the matrix after the infrequent species were deleted but before the relativization.
9. The main matrix should now be relativized by species maximum and contain 47 rows and 103 columns. The contents should be identical to OakWood1.mjm.

Nonmetric multidimensional scaling of the community data, with overlays from the second matrix.

1. Open the file OakWood1.mjm as the main matrix (*File | Open | Main matrix*).
2. Select *Ordination | NMS*.
3. Select the Autopilot tab, check Autopilot mode, and select *Medium*.
4. On the Distance Measure tab, select *Sørensen*. (The selections on the other tabs cannot be set because you have turned on autopilot. Any options previously selected on those tabs will be ignored.)
5. Click **OK**.
6. Enter a descriptive title for the results, for example, "Thilenius data, NMS medium autopilot", then click **OK**.
7. If unsaved results from a previous action are showing in the result window, you will be asked, *Current temporary RESULT.TXT file will be lost. Save file now?* Click on *Discard* or one of the other options, depending on what you want.

- NMS will run, using 40 random starts with the real data set and 50 random starts using different randomizations of the data (shuffling within columns). For each starting configuration NMS will seek a stable 6-, 5-, 4-, 3-, 2-, and 1-dimensional solution.
- When the run is complete, a new result file will appear, along with windows containing coordinates for the stands (GRAPHROW.GPH) and the species (GRAPHCOL.GPH). Save each of these three files with a new name. Select *File* | *Save as* | *Result.txt*, then enter a new name, for example *NMSThil.txt*. Use a similar procedure to save the row and column coordinates, for example as *NMSThil.gph* and *NMSThilSpp.gph*.
- Inspect the result file. See Chapter 16 in McCune & Grace (2002) on NMS. Because random starts are used, your results will differ somewhat from those given here. A key portion of the results file is the following table.

```
STRESS IN RELATION TO DIMENSIONALITY (Number of Axes)
-----
              Stress in real data          Stress in randomized data
              50 run(s)                   Monte Carlo test, 50 runs
-----
Axes  Minimum      Mean      Maximum      Minimum      Mean      Maximum      p
-----
  1    34.496     47.373     56.492     48.881     54.153     56.486     0.0196
  2    22.609     23.775     25.878     29.717     31.645     33.661     0.0196
  3    16.419     16.763     18.039     20.955     22.719     24.182     0.0196
  4    12.319     12.401     12.878     16.829     17.787     18.793     0.0196
-----
p = proportion of randomized runs with stress < or = observed stress
i.e., p = (1 + no. permutations <= observed) / (1 + no. permutations)
```

Conclusion: a 3-dimensional solution is recommended.

- Note that the *p*-values indicate that solutions of any dimensionality from 1 through 4 are stronger than expected by chance. Autopilot chose a 3D solution because it reduces the minimum stress by over 5 units, versus a 2D solution, while still giving a small *p*-value. The final stress for the best 3D solution was 16.4.
- Open the second matrix so you can study the relationship between those variables and the community structure: *File* | *Open* | *Second Matrix* | *OakWood2.mjm*.
- View the ordination graph in 2D. Select *Graph* | *Graph Ordination* | *2D*. See Chapters 13 and 16 in McCune & Grace (2002) for suggestions on how to interpret the results. For example you might wish to:
 - display a joint plot (*Graph* | *Joint Plot*), and examine each pair of axes,
 - see how much of the variation in the distance matrix is represented in the ordination diagram (*Statistics* | *Percent of Variance in Distance Matrix*),

Coefficients of determination for the correlations between ordination distances and distances in the original *n*-dimensional space:

```

      R Squared
Axis  Increment  Cumulative
  1      .221      .221
  2      .282      .502
  3      .246      .749
```

Increment and cumulative R-squared were adjusted for any lack of orthogonality of axes.

- graphically examine the relationships between the ordination and individual variables in the second matrix (*Graph* | *Overlay From Second Matrix*),
 - calculate linear and rank correlation coefficients between axis scores and variables in the second matrix (*Statistics* | *Correlations With Second Matrix*),
 - rotate the diagram so that major vectors are aligned with the axes (select *Graph* | *Joint plot*, then *Rotate* | *By Angle Continuous*. Select 5 degrees for the increment and click *Next* repeatedly to gradually rotate the ordination. See Chapter 15 in McCune and Grace (2002). If you wish to save your rotation, select *File* | *Save Scores As* | *Rows: Stands* | *Text File*, then choose a filename, such as *NMSThilRot.gph*.
 - explore the options – there is a lot here. Take the time to familiarize yourself with the various menu items and options.
- View the ordination graph in 3D. Select *Graph* | *Graph Ordination* | *3D*. The 3D structure is most apparent if you continuously rotate the graph:

First set some formatting options (*Options* | *Preferences*).  We recommend:

- Turn off "Plot points, stands"
- Continuous rotation delay of 40 milliseconds or less.
- Turn off "Continuous rotation labelling"
- Turn on "Label one axis side"

Now click the button with a red circle to rotate continuously.  Click it again to stop.

You can save the rotating figure as an animated GIF, for use in presentations and web sites. To do this, select *File* | *Save Animated GIF As*.

Comparison of communities among groups of stands as defined by categorical variables.

- Open the file *OakWood1.mjm* as the main matrix (*File* | *Open* | *Main matrix*).
- Open the second matrix because it contains the grouping variables: *File* | *Open* | *Second Matrix* | *OakWood2.mjm*.
- Select *Groups* | *MRPP* | *Define Classes From Secondary Matrix*.
- Select one of the categorical variables, for example "Landform" as the grouping variable.
- In the MRPP Setup dialog, select the *Sørensen* distance measure, *n/sum(n)* as the weighting method, and check the box, *Rank transform distance matrix*.
- Click *OK*.
- Enter a descriptive title for the results, for example, "Thilenius data, rank-transformed MRPP, Sorensen distance", then click *OK*.
- If unsaved results from a previous action are showing in the results window, you will be asked, *Current temporary RESULT.TXT file will be lost. Save file now?* Click on *Discard* or one of the other options, depending on what you want.
- MRPP then runs, writing the results into *RESULT.TXT*.
- Inspecting the results window, you should see the following.

```
Chance-corrected within-group agreement, A = 0.07630772
Probability of a smaller or equal delta, p = 0.00772772
```

This A value tells us that species composition rather modestly among landforms, while the *p*-value tells us that it is fairly unlikely for us to obtain this result just by chance. For more help interpreting the results from MRPP, see Chapter 24 in McCune and Grace (2002).

- If you wish to see which species underlie the differences among landforms, try Indicator Species Analysis. From the main menu, select *Groups* | *Indicator Species Analysis* | *Define Classes From Secondary Matrix*.
- Again, select *Landform* as the grouping variable.
- In the Setup dialog, uncheck *Exclude one or more groups* and check *Randomization test*.
- For the random numbers, select *Use time of day*. Click *OK*.
- Select *Number of runs = 1000*, then *OK*.
- Enter a descriptive title for your results, to be used as a header in the results file, then click *OK*.
- If unsaved results from a previous action are showing in the results window, you will be asked, *Current temporary RESULT.TXT file will be lost. Save file now?* Click on *Discard*.
- The results from ISA will appear in the results window. See Chapter 25 in McCune & Grace (2002) for help in interpreting the results.
- One particularly good indicator of landform is "Arel" (*Arrhenatherum elatius*), which has the following result in the final table:

```
-----
              IV from
              Observed randomized
              Indicator groups
Column  Maxgrp Value (IV) Mean S.Dev  p *
-----
  15 Arel      1    86.5   16.6  11.47  0.0050
```

The Maxgrp=1 tells us that it is particularly consistent in Landform 1 (valley bottom). Looking back at the preceding tables, we see that it has a high concentration of abundance in group 1 (86% in the Relative Abundance table) and it was always present in that group (100% in the Relative Frequency table). Your *p*-value may not be exactly the same as this, because you probably had a different sequence of random numbers for the randomization test.

Acknowledgments

The original raw data cards from Thilenius' study in 1963 (data collected in 1961 and 1962) were obtained from John Thilenius via Bob Frenkel. Thanks to John Thilenius for granting permission to distribute his data. Bill Daly did the initial data entry. Bibit Traut added more variables and resolved numerous nomenclatural questions regarding the species codes used by Thilenius.

References

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